



ROOT CAUSE UNLOCKED · CONDITION KIT

The ADHD Root Cause *Kit*

A patient guide to every marker category on the Neural Zoomer Plus and the Organic Acids Test, and what each finding actually changes about your plan

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Nothing in this guide is a reason to stop, skip, or reduce a prescription. If you are taking a stimulant, an antidepressant, or anything else for attention, mood, or sleep, every decision about that prescription belongs with the physician who wrote it. What is in here is meant to sit alongside that care, never to replace it.

Some symptoms need same-day medical attention, not a lab plan. Sudden confusion, sudden severe headache, one-sided weakness or facial droop, trouble speaking, a first seizure, new memory loss after a head injury, or hallucinations are emergencies, and no panel result changes that.

What you are holding, stated plainly

You paid for two panels, and you deserve an honest account of what they are before you read a single result.

The **Neural Zoomer Plus** measures antibody reactivity against neurological antigens. The **Organic Acids Test**, the OAT, measures urinary metabolites left over from cellular metabolism, microbial activity, and neurotransmitter processing. Both are real laboratory measurements. Neither is a diagnostic test for ADHD, for brain fog, or for anything else, and I am not going to pretend otherwise.

Here is the evidence picture as I read it. The individual mechanisms these panels point at are well described in laboratory work: antibodies can interfere with receptor function, gut organisms do produce metabolites that interfere with dopamine synthesis, and inflammation does divert tryptophan away from serotonin. What does not exist is a body of human outcome trials showing that treating people on the basis of these two panels produces better attention or better cognition than treating them without the panels. That trial has not been run, and nobody is lining up to fund it, because most of what these results lead to is unpatentable.

So this is reasoned clinical practice, not settled trial evidence. What I do with that is straightforward. I read these panels as **suggestive rather than diagnostic**. I use them to generate a short list of upstream drivers worth investigating, I sequence the work so the most fixable thing gets addressed first, and then I hold the whole thing to an endpoint, which is your own symptom log compared against your own baseline on a retest. If the markers move and you do not feel better, the markers were not the story. That is the honest standard, and it is the one I apply to my own patients.

One more caution that applies specifically to the OAT neurotransmitter section. Those metabolites are collected in urine, which is outside your brain. Most of your serotonin is made in the gut, and urinary catecholamine metabolites reflect adrenal, gut, and peripheral nerve activity in addition to central activity. A urinary metabolite is a data point about your whole body's chemistry, and it should never be sold to you as a direct readout of your brain chemistry. I read it as one input among several, weighted by whether it matches how you actually feel.

How this guide is organized

Each section tells you what the marker is, what an elevation or a depletion means, and what I actually do about it.

Where I name support, it comes in two tiers.

Tier 1 is what you can start on your own: dietary changes and over-the-counter nutrients.

Tier 2 is targeted support I add in a clinical context, sequenced against your results. I hold no inventory of any kind. I keep a screened dispensary through Fullscript, which ships directly to you, at us.fullscript.com/welcome/drseay. Root Health earns a portion of dispensary purchases, and you should know that when you read any product recommendation I make.

Section 1: Before you read your results

Four rules govern how I read these two panels together. They matter more than any single marker on either report.

Rule 1: The Neural Zoomer Plus reads patterns, not individual markers

A single elevated antibody marker, particularly one in the borderline range, carries much less clinical weight than three or four co-elevated markers across different categories. When blood-brain barrier markers co-elevate with dopamine receptor antibodies and viral infection markers, the pattern points toward a specific proposed mechanism: a prior viral infection breached the barrier and triggered immune reactivity against neurological receptors. That pattern guides what I do next in a way that one isolated marker never does.

Rule 2: Borderline results with a matching symptom picture are clinically meaningful

The Neural Zoomer Plus positive cutoff is the 97.5th percentile. Borderline is defined as 5 percent below that cutoff, which is the 92.5 to 97.5 percentile band. Borderline results in several categories at once are not reassuring. They suggest sub-threshold immune reactivity that may be active and relevant, particularly when your symptom picture is severe or longstanding. I always read a borderline result against the full clinical presentation rather than against the cutoff alone.

Rule 3: Drainage precedes all targeted intervention

The drainage foundation runs for a minimum of two weeks before any targeted neurological, antimicrobial, or antifungal support is introduced. The reasoning is mechanical rather than mystical. Addressing brain inflammation and toxin load mobilizes waste into the lymphatic system and the liver. Those routes need to be

open and working before targeted treatment starts, or the mobilized burden recirculates and you feel worse. I apply this without exception.

Rule 4: The OAT reflects your current biochemical state, not a permanent deficit

Organic acids are urinary metabolites, and they change as the drivers behind them change. A low HVA does not mean your dopamine will always be low. It means that right now, dopamine production or metabolism is impaired, and there are usually upstream reasons for that impairment visible on the same panel. The goal is to find and address those reasons. The metabolite level is the result, not the disease.

Drainage first: the non-negotiable starting point

Tier 1 drainage. Dry brushing for 5 to 10 minutes before your shower, with strokes toward the heart. Castor oil packs over the liver 3 to 4 times weekly. Filtered water at half your body weight in ounces daily. Light rebounding for 10 minutes daily.

Tier 2 drainage. Three products, run for a minimum of 14 days before any targeted support begins:

Product	Clinical role	Dosing
Lymph-Tone II (Energetix)	Lymphatic and extracellular matrix drainage. The right selection for the majority of chronic cognitive cases, where matrix congestion has built up over years. Taken as drops in warm water.	Per label direction
Methylated B12, liposomal or sublingual, taken alongside folate, B6, and TMG (trimethylglycine)	Methylation support for Phase 2 liver detoxification, and direct support for myelin synthesis. B12 does not do this work alone, which is why I run it with its partner nutrients folate, B6, and TMG. Homocysteine Supreme (Designs for Health) is the closest single formula to that combination. Sublingual delivery bypasses the gut for anyone whose absorption is compromised.	2000 mcg daily
ReHydration (Energetix)	Electrolyte and mineral foundation for cellular hydration. Intracellular dehydration is one of the most consistent drainage bottlenecks I see.	Per label direction

Lymph-Tone selection follows the presentation: **Lymph-Tone I** for acute presentations, **Lymph-Tone II** for matrix congestion and most chronic cases, **Lymph-Tone III** for deep or longstanding cases. Each Lymph-Tone is run individually and sequentially, one formula at a time, with no overlap between them. All three are available through my Fullscript dispensary, wherever you are located.

A note on what these are, because I would rather you hear it from me. The Lymph-Tone formulas and ReHydration are homeopathic preparations. The research base does not support homeopathy by the standard I apply to nutrients and botanicals elsewhere in this guide, and I am not going to tell you otherwise. I use them anyway, because in my clinical observation patients tolerate the mobilization phase better when the drainage foundation is in place first, and because the cost and the risk are both low. That is clinical observation, not trial evidence, and I want the label on it to be accurate. Remedy selection is individual enough that it is worth a conversation rather than a guess.

How to read the Neural Zoomer result scale

Negative, below the 92.5th percentile. Below threshold. No antibody production detected against this antigen.

Borderline or moderate, the 92.5 to 97.5 percentile band. Sub-threshold reactivity. Clinically meaningful when several markers are borderline at once, or when the symptom picture is strong.

Positive or high risk, above the 97.5th percentile. Antibody production against this antigen above the laboratory's positive cutoff. This is where I consider protocol-level intervention.

Section 2: What the Neural Zoomer Plus measures

This panel measures antibody reactivity to a set of neurological antigens, organized into eight functional categories. Each category represents a different neurological system. Which category is elevated tells you which system the immune response is directed at. The panel measures IgG, IgA, and IgM subclasses, which gives some indication of whether the response is chronic (IgG dominant), acute mucosal (IgA), or recent (IgM dominant).

Category	Antigen count	Primary relevance for ADHD and brain fog
Blood-Brain Barrier Disruption	4 markers	The most immediately actionable category. Barrier compromise allows other inflammatory and toxic drivers to reach the brain directly.
Brain Inflammation	11 markers	Receptor antibodies that may directly impair neurotransmitter signaling. The proposed explanation for why stimulant medication underperforms in some people despite adequate dopamine availability.
Brain Autoimmunity	13 markers	Broader reactivity against brain tissue, neurotransmitters, and signaling proteins. Often associated with gut permeability, mold toxins, or molecular mimicry with infections.
Demyelination	6 markers	Myelin sheath integrity. Demyelination slows signal conduction, which fits the processing speed deficits, sensory hypersensitivity, and fog people describe.
Optical and Autonomic Nervous System	4 markers	Autonomic dysfunction, sensory processing dysregulation, cranial nerve involvement. Less common as an isolated ADHD finding, important when sensory hypersensitivity and autonomic symptoms are prominent.
Peripheral Neuropathy	5 markers	Peripheral nerve immune reactivity. Elevated alongside barrier markers, it suggests a systemic process affecting both central and peripheral systems.
Neuromuscular Disorders	5 markers	Neuromuscular junction and ion channel antibodies. Less commonly elevated in cognitive presentations, important when muscle fatigue and weakness accompany the cognitive symptoms.
Neurological Infections	7 markers	Viral triggers for neuroinflammation and barrier disruption. Reactivated herpes viruses are among the most consistent upstream findings I see in acquired cognitive dysfunction.

Section 3: Blood-brain barrier disruption markers

The blood-brain barrier is a selective barrier formed by specialized endothelial cells lining the brain's blood vessels. Its job is to regulate what gets in from general circulation. In a healthy state it lets nutrients and oxygen through while blocking pathogens, inflammatory molecules, endotoxins, and neurotoxins. When it is compromised, the brain is exposed to whatever is circulating in the rest of you.

This is the single most important category on the Neural Zoomer Plus for anyone with brain fog or cognitive impairment, because it is the proposed route through which every other upstream driver, whether gut-derived LPS, mold toxins, heavy metals, or infectious organisms, reaches the brain. The four barrier markers each reflect a different aspect of that compromise.

Marker	Function	Elevated suggests	Borderline	First action
Anti-S100B	S100B is a calcium-binding astrocytic protein and a structural component of the barrier. It should not appear in general circulation under normal conditions.	Consistent with a breach of the barrier. These antibodies develop when a brain-specific protein leaks into the bloodstream after barrier damage. Associated with concussion, mold toxins, and chronic neuroinflammation.	Sub-threshold. Worth monitoring if the barrier is a clinical concern.	Evaluate for mold toxin burden. Assess gut permeability. Screen for tick-borne co-infections where history supports it.
Anti-Microglia	Microglia are the brain's resident immune cells. They monitor for threats and initiate inflammatory responses.	Consistent with active immune activation inside the central nervous system, with microglia themselves being targeted. Seen with tissue destruction, toxin exposure, and autoimmune processes.	Sub-threshold.	Address the upstream driver, whether mold, LPS from gut permeability, or tick-borne infection. Add a bioavailable curcumin for NF-kB suppression.
Anti-GFAP	Glial fibrillary acidic protein is expressed by astrocytes, the brain's support cells. Its release into circulation indicates astrocytic injury.	Astrocyte damage. GFAP is released when astrocytes are injured by neurotoxins, inflammatory cytokines, or immune attack. Associated with traumatic brain injury, mycotoxin exposure, and autoimmune encephalopathy.	Sub-threshold.	Same upstream investigation as Anti-S100B. With a head trauma history, assess reverse T3 and the cortisol pattern across the day, since trauma activates the HPA axis.
Anti-GRP78 (glucose regulated protein 78)	GRP78 is an endoplasmic reticulum chaperone protein that manages cellular stress responses. It appears at the cell surface under metabolic or toxic stress.	Endoplasmic reticulum stress in neurons, which is induced by mycotoxins, heavy metals, protein misfolding, and chronic oxidative stress. Suggests significant neuronal metabolic burden.	Sub-threshold.	Liposomal glutathione for ER stress, with N-acetyl cysteine as the glutathione precursor. Address mycotoxin burden where testing supports it.

If any barrier marker is elevated

Investigate the gut connection immediately. Gut permeability drives barrier permeability through LPS endotoxemia. Order a Wheat Zoomer if you have not already. Gut barrier repair runs concurrently with neurological support, not after it.

Test for mold toxin burden. Ochratoxin A specifically increases barrier permeability and inhibits dopamine synthesis. Water-damaged building exposure is the most common source, and it is the one people most often do not think to mention.

Test for tick-borne co-infections if the history is there. *Borrelia* and *Bartonella* both compromise the barrier.

Remove alcohol completely. Alcohol is directly toxic to the endothelial cells that form the barrier. This one is free and it works faster than anything else on the list.

Tier 1 support for elevated barrier markers

- **Lion's mane mushroom extract, 500 to 1000 mg daily.** Supports nerve growth factor production, which is the proposed route to barrier repair and remyelination.
- **Omega-3 fatty acids, EPA plus DHA at 2000 to 3000 mg daily, triglyceride form.** Reduces neuroinflammation and supports barrier membrane integrity through phospholipid composition.
- **Phosphatidylserine, 100 mg three times daily.** Supports neuronal membrane repair and blunts cortisol-driven barrier compromise.
- **Complete gluten elimination.** Gliadin antibodies cross-react with neurological tissue.
- **Remove seed oils** (canola, soybean, corn, sunflower). Omega-6 overload amplifies inflammatory signaling.

Tier 2 support for elevated barrier markers

- **A bioavailable curcumin preparation**, per label direction, for upstream NF-kB and COX-2 suppression. The catch with curcumin is bioavailability, so the preparation matters more than the milligram count.
- **A broad-spectrum binder**, per label direction, where mold or heavy metal burden is documented alongside barrier elevation. Binders go **1 to 2 hours away from food, supplements, and medication**, without exception, or they will bind the things you are trying to absorb.
- **MycoCan-Chord (Energetix)**, per label direction, the homeopathic mold and fungal formula I pair with the binder when mold burden is documented. Like the Lymph-Tone remedies in the drainage section, this is a homeopathic preparation, and the same honest label applies: clinical observation, not trial evidence.
- **Targeted astrocyte and glymphatic support**, where indicated. The original version of this protocol used an energetic remedy in this role, a paradigm in which any comparable spagyric line substitutes freely, so no single label is worth a premium over another. If what you want is the outcome that kind of product

claims, meaning memory, brain repair, and glymphatic support, the dose-response evidence belongs to citicoline, phosphatidylserine, bacopa, magnesium L-threonate, and a DHA-weighted omega-3 instead. I present both honestly and let you choose the paradigm. For my current pick on the energetic side, ask me in the Discovery Session.

Section 4: Brain inflammation markers

This is the category most directly relevant to ADHD and cognitive dysfunction. These 11 markers include antibodies against the neurotransmitter receptors themselves. Laboratory work indicates that when such antibodies are present they are not passive bystanders: they interfere with receptor function, which would produce a functional neurotransmitter deficit operating independently of how much neurotransmitter is actually being made. That is the proposed mechanism behind incomplete stimulant response in some people. The medication raises dopamine availability, but the signal is impaired at the receptor.

Of everything on this panel, this category is the most likely to be oversold, so I will go carefully. The receptor antibody findings are real measurements and the mechanism is plausible and supported in laboratory work. What has not been demonstrated is that lowering these antibodies improves attention in a controlled trial. I treat them as a reason to go looking upstream, not as a diagnosis in themselves.

Dopamine receptor antibodies: Anti-DR1 and Anti-DR2

Dopamine receptor 1 sits primarily on post-synaptic neurons and mediates the stimulating effects of dopamine, including reward, motivation, reinforcement learning, and executive function. Dopamine receptor 2 is expressed on both pre-synaptic and post-synaptic neurons, where it mediates inhibitory modulation of dopamine release, along with attention, working memory, and impulse control. Between them, DR1 and DR2 govern most of the functions that are impaired in ADHD.

These are the two most clinically important markers in this category for attention and fog presentations. When elevated, the proposed effect is direct impairment of dopamine signaling at the receptor, which would maintain an ADHD-like state regardless of how much dopamine is produced or prescribed. These antibodies have documented cross-reactivity with Streptococcal A antigens through molecular mimicry, so an elevation here should always send you to the Anti-Streptococcal A result in the infections section of the same panel.

NMDA and AMPA receptor antibodies

The NMDA receptor is the primary mediator of learning, memory formation, and synaptic plasticity. It is a glutamate receptor operating a calcium channel, and its function is essential to long-term potentiation, the cellular mechanism behind memory consolidation. Anti-NMDA antibodies impair memory formation, learning speed, and processing. In severe cases, anti-NMDA encephalitis is a recognized autoimmune syndrome causing psychosis, seizures, and severe cognitive impairment. Milder, earlier-stage reactivity is proposed to produce the memory difficulty and fog that many people with attention complaints describe.

The AMPA receptor mediates fast excitatory synaptic transmission. Antibodies against it impair the speed and reliability of synaptic signaling, which fits slow processing speed, inconsistent cognitive performance, and difficulty sustaining focus on demanding tasks.

GABA and glycine receptor antibodies

GABA is the primary inhibitory neurotransmitter, and its job is to modulate and calm neurological activity. Anti-GABA receptor antibodies impair the brain's ability to regulate excitatory signaling, producing chronic hyperactivation. It shows up as anxiety, hyperexcitability, sensory hypersensitivity, insomnia, and the restless physical and mental activity characteristic of ADHD. Anti-glycine receptor antibodies impair a second inhibitory pathway and are associated with exaggerated startle response, rigidity, and difficulty with sensory regulation.

The remaining brain inflammation markers

Marker	What elevation means	Pattern note
Anti-DPPX (dipeptidyl aminopeptidase-like protein 6)	DPPX regulates voltage-gated potassium channels in the brain. Antibodies impair that regulation, producing hyperexcitability, agitation, and GI symptoms alongside cognitive changes.	Associated with VGKC complex autoimmunity. Read alongside Anti-VGKC in the neuromuscular section.
Anti-Neurexin 3	Neurexin 3 is a synaptic organizing protein essential to synapse formation and function. Antibodies disrupt transmission between neurons.	Suggests established synaptic dysfunction. Often co-elevated with NMDA or AMPA.
Anti-CASPR2 (contactin associated protein-like 2)	CASPR2 regulates voltage-gated potassium channels at the nodes of Ranvier. Antibodies produce limbic encephalitis features (memory impairment, psychiatric symptoms, insomnia, and autonomic dysfunction).	Strong association with HHV-6 and other viral triggers. Check the infections section.
Anti-LGI1 (leucine-rich glioma-inactivated 1)	LGI1 modulates AMPA receptor function and is highly expressed in the hippocampus. Antibodies are associated with faciobrachial dystonic seizures, severe amnesia, and psychiatric symptoms.	Less common in attention presentations. More significant in encephalopathy or seizure contexts.
Anti-Ma	Ma proteins are expressed in neurons and testes. These antibodies are associated with limbic and brainstem encephalitis, producing memory loss, sleep disorders, and cognitive decline.	Elevation warrants investigation for paraneoplastic syndrome where symptom severity is extreme. This one goes to a physician.

Tier 1 support for brain inflammation markers

- **Magnesium glycinate or threonate, 300 to 400 mg before bed.** Magnesium is an NMDA receptor antagonist and a GABA cofactor. Deficiency is extremely common and it directly amplifies excitotoxicity.
- **Zinc bisglycinate, 15 to 30 mg with food.** A direct modulator of dopamine metabolism and NMDA receptor function.
- **Remove artificial food dyes** (Red 40, Yellow 5, Yellow 6) **and sodium benzoate.** These have documented effects on dopamine receptor function and hyperactivity, and are legally required to carry warnings in the EU.
- **Address gut permeability.** LPS from a leaky gut is a primary proposed upstream driver of anti-dopamine receptor antibody production.

Tier 2 support for brain inflammation

- **Baicalin (Chinese skullcap),** per label direction, for neuroprotective NF-kB suppression with specific activity against inflammatory cytokine production in central nervous system tissue.
- **Liposomal zinc** for receptor and dopamine support, per label direction.
- **Low-dose lithium orotate** for neuroprotection and BDNF support. This is a micro-dose, nutritional-range mineral taken per label direction, nothing close to pharmaceutical lithium dosing. If you take any psychiatric medication, raise this with your prescriber before starting it.

Section 5: Brain autoimmunity markers

This category covers 13 markers against brain tissue proteins, neurotransmitter molecules, and signaling receptors. It is distinct from brain inflammation in that it reflects broader reactivity against brain components rather than specific receptor antibodies. Several of these markers relate to neurodegenerative processes and to autoimmune neurological conditions that can present as cognitive dysfunction or ADHD-like symptoms in adults.

Anti-dopamine and anti-hydroxytryptamine (5-HT)

Anti-dopamine antibodies target the dopamine molecule itself, not its receptor. Their presence suggests the immune system is treating dopamine as an antigen, which would deplete it through a route entirely separate from synthesis capacity. When I see anti-dopamine antibodies together with a low HVA on the OAT, the depletion has two concurrent proposed causes: reduced production and immune-mediated destruction. In my experience this combination is the most stubborn and most medication-resistant dopamine picture on the panel.

Anti-hydroxytryptamine antibodies target serotonin. Elevation here fits the mood instability, emotional dysregulation, rejection sensitivity, and sleep disruption that so often ride alongside the attention symptoms. These are associated with gut permeability, which makes sense given that the gut produces more than 90 percent of the body's serotonin. A leaky gut allows serotonin and its precursors to become immunogenic.

Anti-glutamate

Glutamate is the primary excitatory neurotransmitter. These antibodies are proposed to create a paradoxical state: blocking normal glutamate receptor function while simultaneously triggering inflammatory cytokine cascades that produce excess excitation. The clinical picture is cognitive impairment, anxiety, attention difficulty, and sleep disruption that does not respond predictably to either stimulant or calming interventions. When someone tells me that everything they try backfires, this is one of the markers I want to see.

Amyloid beta and tau markers

Anti-amyloid beta (1 to 42) and anti-amyloid beta (25 to 35) antibodies reflect immune reactivity against amyloid beta peptides, which are associated with neurodegeneration when they accumulate in plaques. In a younger person with cognitive complaints, elevation is much less likely to reflect established Alzheimer's pathology and much more likely to reflect amyloid beta-driven neuroinflammation related to toxin burden, barrier disruption, or metabolic stress. Anti-tau antibodies reflect reactivity against the tau protein, which forms neurofibrillary tangles in neurodegenerative conditions but also appears as an inflammatory marker at early and sub-clinical stages.

Elevation of amyloid beta or tau markers alongside significant barrier disruption and mold toxin burden is the combination I treat most urgently. Ochratoxin A has documented associations with tau phosphorylation and amyloid beta accumulation. When those co-elevate, the mold work becomes the immediate priority and everything else waits.

Anti-alpha-synuclein

Alpha-synuclein is a protein that aggregates into Lewy bodies in Parkinson's disease and Lewy body dementia. These antibodies in the context of cognitive complaints, executive dysfunction, and movement symptoms warrant real clinical attention. In early presentations, alpha-synuclein reactivity is most commonly associated with gut microbiome dysbiosis through the gut-brain axis, with pesticide or toxin exposure, particularly organophosphates and heavy metals, and with mitochondrial dysfunction. The gut-origin hypothesis for alpha-synuclein pathology has substantial support in the current literature and is still argued, which is the accurate way to state it.

The remaining brain autoimmunity markers

Marker	Clinical significance	Action
Anti-Cerebellum	Cerebellar autoimmunity is associated with balance problems, coordination difficulty, gait disturbance, and handwriting changes. In attention presentations it may explain fine motor and sensory integration difficulty.	Assess for gluten cross-reactivity. Wheat Zoomer is indicated. Cerebellar antibodies have a documented association with gluten sensitivity.
Anti-Purkinje Cell	Purkinje cells are the primary output neurons of the cerebellar cortex. Antibodies are associated with paraneoplastic cerebellar degeneration and autoimmune cerebellar ataxia. Isolated elevation in a cognitive presentation warrants attention to underlying autoimmune and infectious triggers.	Warrants a broader autoimmune workup. Consider an ANA IFA panel.
Anti-Yo	A paraneoplastic marker with associations to ovarian and breast cancers in women and lung cancer in others. In isolation it may instead indicate post-viral or autoimmune cerebellar inflammation.	Elevation with neurological symptoms should be discussed with a physician. Do not sit on this one.
Anti-Adrenergic Receptors (alpha1 and beta2)	Adrenergic receptors mediate the sympathetic nervous system response. Antibodies impair autonomic regulation, producing dysautonomia with palpitations, blood pressure instability, orthostatic intolerance, inappropriate tachycardia, and exaggerated stress responses.	Assess the cortisol pattern across a full day rather than from a single morning draw, because a flat-low curve and a wired-high curve both present as exhaustion and call for different approaches. Mast cell activation is a concurrent possibility worth considering.
Anti-Endothelin A Receptor	The endothelin A receptor regulates vascular tone in the brain. Antibodies impair cerebral blood flow regulation, which may contribute to migraine, vasomotor instability, and cognitive performance that fluctuates through the day.	Address cardiovascular inflammatory burden. Bioavailable curcumin and anti-inflammatory dietary changes.
Anti-RAGE Peptide	RAGE, the receptor for advanced glycation end-products, mediates inflammatory signaling in response to metabolic damage, oxidative stress, and glycotoxins. These antibodies indicate elevated glycation burden and are associated with diabetes, metabolic syndrome, and advanced neuroinflammation.	Blood sugar stabilization becomes the priority. Remove refined sugar and refined carbohydrate. Assess fasting glucose, HbA1c, and fasting insulin , because insulin rises years before glucose does and a glucose-only workup will miss the window.

Section 6: Demyelination markers

Myelin is the fatty insulating sheath around nerve axons that enables rapid electrical conduction. It works the way insulation works on electrical wire. Without it, signals slow, cross over, or fail to arrive reliably.

Demyelinating antibodies target the proteins and lipoproteins that make up that sheath. Slowed transmission fits the processing speed deficits, sensory hypersensitivity, and fatigue that are among the most commonly reported complaints in brain fog and treatment-resistant attention problems.

Myelin has documented structural similarity with gliadin, the immunogenic fraction of gluten. Anyone with elevated demyelination antibodies who is still eating gluten may be maintaining the process through ongoing dietary molecular mimicry. I order a Wheat Zoomer as a co-test for every patient with elevated demyelination markers.

Marker	Function	Elevated suggests	Borderline	First action
Anti-MAG (myelin-associated glycoprotein)	MAG maintains the spacing between myelin lamellae and mediates communication between axon and myelin. It is concentrated at the outermost myelin layer.	Anti-MAG neuropathy produces slowly progressive sensory neuropathy with imbalance, distal sensory loss, and hand tremor. In fog presentations, elevation contributes to sensory processing and coordination difficulty.	Sub-threshold.	Wheat Zoomer for molecular mimicry. Remove gluten. Omega-3 fatty acids for myelin membrane support.
Anti-Neurofascin	Neurofascin organizes sodium channels at the nodes of Ranvier and paranodal regions. It is essential to rapid saltatory conduction.	Impaired saltatory conduction, meaning signal transmission slows substantially. Fits processing speed deficits, fatigue with cognitive effort, and focus that varies through the day.	Sub-threshold.	Address infectious triggers. Check HSV, EBV, and HHV-6 in the infections section. B12 and selenium support.
Anti-MOG (myelin oligodendrocyte glycoprotein)	MOG is expressed on the outermost myelin surface and is a target in autoimmune demyelinating disease.	Associated with optic neuritis, transverse myelitis, and encephalomyelitis. At lower levels, suggests active demyelinating autoimmunity that may be driving fog and neurological fatigue.	Sub-threshold.	Elevation warrants neurological evaluation. Address gut, thyroid, and mold as upstream drivers alongside that referral.
Anti-MBP (myelin basic protein)	MBP is one of the most abundant proteins in central nervous system myelin and is involved in myelin compaction and stabilization.	Active central demyelination. MBP antibodies are elevated in multiple sclerosis and also in sub-clinical autoimmune demyelination associated with gut permeability, gluten mimicry, and mold toxins.	Sub-threshold.	Strict gluten elimination. MBP shares molecular mimicry with gliadin. Myelin repair support: lion's mane, phosphatidylcholine.

Marker	Function	Elevated suggests	Borderline	First action
Anti-PLP (proteolipid protein)	PLP is the major protein of central nervous system myelin, at roughly 50 percent of myelin protein content.	Significant central myelin involvement. Less specific than anti-MOG, but reflects broader myelin autoimmunity.	Sub-threshold.	Same approach as anti-MOG. Phosphatidylserine and phosphatidylcholine for membrane repair.
Anti-Tubulin	Tubulin is the primary structural protein of microtubules, which form the intracellular transport network inside neurons.	Impaired axonal transport. Microtubule disruption impairs movement of neurotransmitter vesicles and cellular components along the axon, effectively slowing the neuron from the inside.	Sub-threshold.	Assess heavy metal burden if anti-tubulin is elevated alongside barrier markers. Mercury specifically disrupts tubulin polymerization.

General support for demyelination findings

Tier 1. Lion's mane extract, 500 to 1000 mg daily, for nerve growth factor stimulation supporting Schwann cell activity and myelin maintenance. Omega-3 fatty acids, 2 to 3 g daily, since the myelin membrane is largely phospholipid and DHA dependent. Phosphatidylcholine, 1200 mg daily, as a choline-rich phospholipid for membrane biosynthesis. Complete gluten elimination for the molecular mimicry with MBP and other myelin proteins.

Tier 2. Continue the drainage foundation, including the methylated B12. Add liposomal zinc for enzyme function in myelin synthesis.

Section 7: Neurological infection markers

This section tests for antibodies against seven viral and bacterial organisms known to breach the blood-brain barrier, trigger neuroinflammation, and initiate or sustain neurological autoimmunity. These are not general infectious disease tests. They are specific to organisms with well-documented neurological tropism and with evidence of persisting as drivers of cognitive impairment after the acute infection has resolved.

The key principle is that these viruses do not need to be actively replicating to maintain neurological dysfunction. Reactivated latent herpes viruses in particular can produce meaningful immune activation through periodic subclinical reactivation that generates ongoing inflammatory signaling without ever producing a visible outbreak. This mechanism is underappreciated in conventional practice, and it is one of the most consistent findings I encounter in treatment-resistant brain fog.

Marker	Function	Elevated suggests	Borderline	First action
Anti-HSV1 and Anti-HSV2	HSV1 has strong neurological tropism and establishes latency in the trigeminal ganglia. HSV2 establishes latency primarily in the sacral ganglia but produces neurological complications in immunocompromised states.	Active or reactivated HSV with neurological involvement. HSV1 encephalitis is a recognized cause of severe cognitive impairment, and subclinical HSV1 reactivation is associated with hippocampal inflammation and memory impairment.	Prior seroconversion without active neurological involvement.	Lysine competes with the arginine required for HSV replication. Limit arginine-rich foods including nuts and chocolate. Targeted viral immune support, discussed below.
Anti-EBV (Epstein-Barr virus)	EBV is the primary cause of mononucleosis and establishes lifelong latency in B cells. Its neurological tropism produces encephalitis, cerebellitis, and chronic fatigue with profound cognitive impairment.	EBV reactivation with neurological involvement. This is one of the most common findings I see in post-infectious brain fog, chronic fatigue, and acquired cognitive impairment.	Prior exposure without active reactivation.	Evaluate immune function, and address what permitted reactivation, such as HPA axis dysregulation, mold burden, nutrient depletion, or sleep deprivation.
Anti-CMV (cytomegalovirus)	CMV establishes latency in monocytes and can reactivate in immunocompromised states. When CMV causes encephalitis, it produces cognitive decline, personality changes, and focal deficits.	CMV reactivation with neurological involvement. Less common than EBV in attention and fog presentations, significant when present.	Prior exposure without reactivation.	Same immune approach as EBV. Assess for the immune dysfunction driving reactivation rather than treating the titer.
Anti-HHV-6 and Anti-HHV-7	Both are neurotropic and establish latency in the central nervous system. HHV-6 integrates into chromosomes and can reactivate in the brain	HHV-6 reactivation is among the most consistently reported findings in ME/CFS and fibromyalgia research. In fog and	Prior exposure without active central nervous system involvement.	The full immune reactivation approach. Address mold, the cortisol pattern, and

Marker	Function	Elevated suggests	Borderline	First action
	independently of systemic immune suppression.	attention presentations it is a leading proposed driver of limbic inflammation, memory impairment, and cognitive fatigue.		sleep architecture at the same time, because reactivation tracks with all three.
Anti-Streptococcal A	Group A Streptococcus produces surface proteins with molecular mimicry against dopamine receptors, the basal ganglia, and neurological tissue. This is the mechanism underlying PANDAS.	Streptococcal A antibodies co-elevating with Anti-DR1, Anti-DR2, or Anti-NMDA is consistent with the PANDAS and PANS mechanism, where streptococcal molecular mimicry drives receptor autoimmunity. Relevant in adults as well as children, with OCD, tic disorders, attention difficulty, and emotional dysregulation.	Previous exposure without active molecular mimicry.	With dopamine receptor antibodies, treat the ongoing streptococcal burden directly: berberine and oregano oil. Address gut dysbiosis as the streptococcal reservoir.

Immune support for reactivated viral infections

Tier 1. Zinc, 15 to 30 mg daily. Vitamin D, maintained in the functional target range of 40 to 80 ng/mL. Vitamin C, 1000 mg two to three times daily. Lysine, 1000 mg twice daily on an empty stomach, for the herpes family specifically. And deep sleep optimization, which is not a throwaway line: the immune surveillance that holds viral latency in place happens during deep sleep, so a sleep problem is a viral problem.

Tier 2. Continue the B12 combination from the drainage phase. **Viru-Chord (Energetix)**, per label direction, is the viral immune support formula I use here. It is a homeopathic preparation, and the same honest label from the drainage section applies. The original protocol also paired it with thymic and colostrum-based immune support; I have not settled on the replacements I want to stand behind for those, so ask me for the current products in the Discovery Session rather than guessing from a shelf.

Section 8: The other three Neural Zoomer categories

These remaining categories are less commonly the primary finding in attention and cognitive presentations, but they become important when they co-elevate alongside the categories already covered. An isolated peripheral neuropathy elevation in someone with no neuropathy symptoms carries far less urgency than the same elevation appearing alongside barrier markers and viral markers. Context and pattern determine priority, every time.

Optical and autonomic nervous system markers

Marker	What elevation means	Cognitive relevance
Anti-Neuron Specific Enolase (NSE)	NSE is released from neurons and neuroendocrine cells during injury or death. Elevated antibodies indicate ongoing neuronal damage. NSE is used as a serum marker of brain injury severity in trauma and cardiac arrest.	Elevated NSE alongside barrier markers points to significant neuronal damage. This raises the urgency of identifying the upstream driver.
Anti-Aquaporin 4 (AQP4)	AQP4 regulates water movement across the barrier and is involved in astrocyte function and brain edema. It is the antibody target in neuromyelitis optica spectrum disorder.	Elevation alongside barrier markers suggests astrocytic dysfunction contributing to fog. AQP4 dysfunction impairs glymphatic waste clearance, which is the brain's overnight cleaning system.
Anti-Recoverin	Recoverin is expressed in retinal photoreceptors and some central neurons. These antibodies are associated with cancer-associated retinopathy and paraneoplastic retinal degeneration.	Less common in attention presentations. Isolated elevation warrants ophthalmological evaluation.
Anti-CVX (Coxsackie virus)	Coxsackie virus can produce viral encephalitis and post-viral neurological syndromes.	Relevant when cognitive symptoms followed an acute viral illness. Same immune approach as EBV and CMV.

Peripheral neuropathy markers (Anti-GM1, Anti-GM2, Anti-Hu, Anti-Ri, Anti-Amphiphysin)

These target gangliosides (GM1, GM2) and RNA-binding proteins (Hu, Ri) associated with paraneoplastic neuropathy and immune-mediated peripheral nerve disease. In attention and fog presentations they matter

most when they co-elevate with central markers, since that combination suggests a systemic process affecting the peripheral and central nervous systems at once.

Anti-GM1 and Anti-GM2 target ganglioside lipids in peripheral nerve myelin, and elevation is associated with Guillain-Barre variants, multifocal motor neuropathy, and peripheral neuropathy. Anti-Hu, Anti-Ri, and Anti-Amphiphysin are paraneoplastic antibodies that should be discussed with a physician when elevated, since they have associations with small cell lung cancer and other malignancies, though they also appear in autoimmune neurological conditions. I do not manage those findings alone and neither should anyone else.

Neuromuscular markers (Anti-MuSK, Anti-VGCC, Anti-VGKC, Anti-Titin, Anti-AChR)

These target components of the neuromuscular junction. Anti-muscle specific kinase antibodies are associated with myasthenia gravis. Anti-voltage gated calcium channel antibodies are linked to Lambert-Eaton syndrome and also appear in dysautonomia. Anti-voltage gated potassium channel complex antibodies produce limbic encephalitis characterized by memory impairment, psychiatric symptoms, and seizures. Anti-titin and anti-acetylcholine receptor antibodies are additional myasthenia gravis-associated markers.

In cognitive presentations, the VGKC complex is the finding that matters most. Anti-VGKC elevation produces limbic encephalitis symptoms (impaired memory, behavioral change, anxiety, sleep disturbance, and cognitive decline) that closely mimic severe ADHD and depression. The DPPX antibody from the brain inflammation section is part of the VGKC complex, so read those two results together rather than separately.

Section 9: What the Organic Acids Test measures

The OAT measures urinary metabolites, the byproducts of cellular metabolism, microbial activity, and neurotransmitter processing, which together reflect the biochemical environment the brain is operating in. Where the Neural Zoomer Plus indicates what the immune system is reacting against, the OAT indicates how the chemistry is performing right now.

The panel covers seven categories: microbial markers, bacterial and fungal; neurotransmitter metabolites and ratios; energy metabolism and mitochondrial function, including the Krebs cycle, glycolysis, and ketone and fatty acid oxidation; detoxification and oxidative stress markers; amino acid metabolites; nutrition and oxalate markers; and pyrimidine metabolites.

How to read it. Unlike the Neural Zoomer Plus, the OAT does not use percentile cutoffs. Results are reported against reference ranges, with clinical significance increasing as values move further outside the normal range.

The key principle: do not read the OAT in isolation. The OAT shows you the biochemical consequences of the upstream drivers. Elevated kynurenic acid means tryptophan is being diverted from serotonin into the inflammatory kynurenine pathway, which means there is an upstream inflammatory driver somewhere. The

Neural Zoomer Plus helps identify what that driver is. Reading either panel alone is how people end up treating a metabolite instead of a cause.

Section 10: OAT bacterial markers

These identify specific bacterial organisms and classes by their urinary metabolic byproducts. They are not bacterial counts. They are the chemical signatures certain bacteria leave in metabolic output, and their relevance to cognition is substantial, because specific overgrowths produce metabolites that interfere with dopamine.

HPHPA, the highest-priority bacterial finding

HPHPA, or 3-(3-hydroxyphenyl)-3-hydroxypropionic acid, is the primary metabolite of *Clostridia* species in the gut, particularly *C. sporogenes* and *C. caloritolerans*. *Clostridia* produce dopamine-modifying metabolites that interfere with dopamine synthesis and metabolism. HPHPA elevation is among the most consistently reported findings in autism, ADHD, and schizophrenia research. The proposed effect is direct impairment of the HVA and dopamine pathway, producing a dopamine-deficient picture regardless of how much precursor is consumed or how much stimulant is prescribed.

This is the finding I address first when it is present, because *Clostridia*-driven dopamine impairment will blunt any neurotransmitter precursor support you layer on top of it.

What I do about it. Tier 1: *L. acidophilus* and bifidus probiotic support, since fermented foods increase competition against *Clostridia*. Strict removal of refined sugar and processed food. Botanical antimicrobials with *Clostridia* activity, including oregano oil, allicin from garlic, carvacrol, and baicalin. Tier 2: baicalin specifically for its anti-*Clostridia* activity, alongside gut barrier repair and multi-strain probiotic support for microbiome restoration.

The remaining bacterial markers

Marker	Elevation indicates	Clinical priority
4-Cresol	Produced by <i>C. difficile</i> and other Clostridia through fermentation of tyrosine. Elevation indicates active Clostridia overgrowth with <i>C. difficile</i> involvement. 4-Cresol inhibits dopamine synthesis by blocking dopamine beta-hydroxylase, and it is immunotoxic, suppressing gut mucosal immunity and promoting permeability.	High priority. <i>C. difficile</i> involvement calls for careful antimicrobial selection. Do not introduce broad-spectrum antimicrobials without drainage preparation first. <i>Saccharomyces boulardii</i> is specifically indicated here, since it produces proteases that degrade <i>C. difficile</i> toxins.
DHPPA (3,4-dihydroxyphenylpropionic acid)	Produced by beneficial bacteria, including Lactobacilli, Bifidobacteria, and <i>E. coli</i> , when they ferment dietary polyphenols. A marker of beneficial bacterial metabolism.	Low DHPPA reflects depletion of beneficial bacteria, low polyphenol intake, or both. High DHPPA reflects increased beneficial activity or polyphenol excess. Low is the more common and more concerning finding in this population.
Hippuric Acid	Formed when glycine conjugates with benzoic acid, which can come from dietary preservatives, bacterial metabolism of polyphenols, or environmental toluene exposure.	Low: impaired glycine availability or low polyphenol metabolism. High: chemical and preservative exposure such as sodium benzoate in processed food, toluene exposure, or specific overgrowths.
4-Hydroxybenzoic Acid, 4-Hydroxyhippuric Acid	Related metabolites of benzoic acid and phenolic compound metabolism. They are also enzyme conversion products of elevated HPPA.	Elevated alongside HPPA, they are consistent with active Clostridia-driven disruption of the dopamine pathway. Address the HPPA finding.
4-Hydroxyphenylacetic Acid	Produced by specific bacterial species. Elevated levels reflect small bowel bacterial overgrowth patterns, particularly in the upper GI tract.	Suggests SIBO or a specific dysbiotic overgrowth. Consider a GI-MAP or Gut Zoomer for full characterization.
Indoleacetic Acid	A tryptophan metabolite produced by Lactobacillus and other commensals. It acts as a gut immune modulator and an aryl hydrocarbon receptor ligand.	Low: beneficial bacterial depletion. High: specific overgrowths or excess tryptophan fermentation. Interpretation depends on the concurrent tryptophan metabolites (kynurenic acid, quinolinic acid, and 5-HIAA).

Section 11: OAT yeast and fungal markers

These identify Candida overgrowth and Aspergillus colonization by their urinary metabolic signatures. Fungal overgrowth is a clinically underappreciated contributor to cognitive dysfunction. Candida produces acetaldehyde, which interferes with dopamine synthesis, competes with tryptophan for transport, and induces mitochondrial dysfunction. Aspergillus species produce multiple mycotoxins with documented neurological toxicity.

This matters practically: OAT fungal markers reflect organisms in the small intestine and colon, which standard endoscopy and conventional stool culture do not reach. If you have persistent gut-related cognitive symptoms and a completely negative standard GI workup, this section is often where the explanation finally appears.

Marker	Elevation indicates	Clinical priority
Arabinose	The primary <i>Candida albicans</i> marker. A sugar released by Candida during metabolic activity. Elevated urinary arabinose indicates active Candida overgrowth in the GI tract. Arabinose has been reported to fall following antifungal therapy, which is why I use it as a retest marker rather than a one-time reading.	The highest-priority fungal marker. Elevation calls for antifungal dietary intervention, meaning sugar out completely, plus targeted botanical antifungal support, before any neurotransmitter precursor supplementation.
Tartaric Acid	Produced by yeast and fungal species, and also a Krebs cycle inhibitor that blocks fumarase activity directly. Elevated tartaric acid impairs cellular energy production independently of any mitochondrial pathology.	Active fungal overgrowth with concurrent energy impairment. Address the fungal overgrowth before introducing mitochondrial support, because the mitochondrial markers will not normalize while the fumarase blockade continues. This is a sequencing error I see constantly.
3-Oxoglutaric Acid	An Aspergillus-associated marker reflecting small bowel fungal colonization. Co-elevates with other Aspergillus markers where there is a mold exposure history.	Aspergillus colonization, correlating with mold toxin exposure. Order a Vibrant Mycotoxins panel if this is elevated alongside any barrier or neurological markers.
Citramalic Acid	Produced by Candida and other yeast species. Less specific than arabinose.	Moderately elevated: supports the Candida picture when arabinose is also elevated. Isolated elevation is less clinically significant.
5-Hydroxymethylfuroic Acid, Furan-2,5-dicarboxylic Acid, Furancarboxylglycine	These three furanic compounds are specific Aspergillus metabolites and are the OAT markers most directly related to building mold exposure. They are distinct from food-derived arabinose elevation and reflect actual colonization or residual toxin burden from environmental exposure.	If furanic compounds are elevated alongside any brain inflammation or barrier markers, order the Mycotoxins panel. These markers are consistent with the biological pathway by which mold toxins reach the brain.
Tricarballic Acid	A Fusarium-derived metabolite. Fusarium is a common soil and food mold found on grains, on corn, and in water-damaged buildings. Tricarballic acid inhibits aconitase, a critical Krebs	Indicates Fusarium exposure or colonization. Correlates with fatigue, post-exertional malaise, and exercise intolerance alongside the cognitive symptoms.

Marker	Elevation indicates	Clinical priority
	cycle enzyme, directly impairing energy production.	
Carboxycitric Acid	An intestinal yeast and fungal overgrowth marker with limited but suggestive evidence. Also a Krebs cycle byproduct.	Moderately elevated: supports the fungal picture, best read alongside the other fungal markers rather than alone. Low: potentially decreased Krebs cycle activity.

If any fungal markers are elevated

Eliminate refined sugar and high-fructose corn syrup completely. Candida and Aspergillus are selective consumers of simple sugars, and sugar elimination deprives the organism of its primary growth substrate.

Remove alcohol. Alcohol is metabolized to acetaldehyde, the same metabolite Candida produces. Drinking while carrying a Candida overgrowth doubles the acetaldehyde burden.

Consider the Mycotoxins panel if the Aspergillus markers, meaning the furanics and 3-oxoglutaric acid, are elevated alongside any barrier or neurological markers. The Aspergillus OAT markers do not rule out environmental mycotoxin exposure, and may be the biological signature of it.

Tier 2 antifungal support

MycoCan-Chord (Energetix), per label direction, is the homeopathic fungal drainage formula I stage first, before any botanical antifungal begins. After that come two botanical antifungals with documented Candida activity. **Pau d'arco** is taken per label direction, and **oregano oil** is capped at **1 to 2 capsules total per day**. That cap is deliberate. Oregano oil is potent, and building past that total is where I see people get into trouble. **Saccharomyces boulardii**, per label direction, is a beneficial yeast that competes with Candida for mucosal binding sites and stimulates secretory IgA production.

Begin antifungal support only after drainage is established. Candida die-off produces significant acetaldehyde and endotoxin burden, and that burden has to be processed through an open drainage system.

A word on feeling worse, because you may. Some people feel rougher for a stretch after starting antifungal work. How you read that matters, because a version of this framing floating around cannot be argued with. If feeling better proves it is working and feeling worse proves it too, the framework is unfalsifiable and it is not doing you any favors. Worsening is information. Sometimes it passes in a few days. Sometimes it means the dose is too high, or the drainage groundwork was rushed. Significant worsening means we reassess, not that we push through. Reach out rather than gutting it out.

Section 12: OAT neurotransmitter markers and ratios

This is the most directly actionable OAT section for attention and fog presentations. These markers reflect production and breakdown of dopamine and serotonin. Read the caution from the opening of this guide alongside them: these are urinary metabolites collected outside the brain, and they are suggestive rather than diagnostic. I weight them by whether the pattern matches how you actually feel and function.

Catecholamine metabolites, the dopamine pathway

The dopamine pathway produces three measurable metabolites (DOPAC, HVA, and VMA). Their relative levels and two calculated ratios characterize dopamine metabolism far better than any single marker.

Marker	What it reflects	Interpretation
HVA (homovanillic acid)	The primary end metabolite of dopamine catabolism, reflecting total dopamine production and turnover. Produced when dopamine is broken down by MAO and COMT in the synapse.	Low HVA points to dopamine depletion or impaired synthesis. The common associations are Clostridia HPHPA, Candida acetaldehyde, nutrient shortfalls in tyrosine, B6, iron, or copper, and mold toxin burden. High HVA signals excessive turnover, often stress-driven, with rapid depletion.
DOPAC (3,4-dihydroxyphenylacetic acid)	An intermediate dopamine metabolite reflecting intraneuronal breakdown via MAO. It reports on the balance between synthesis and breakdown inside the neuron.	Low DOPAC with low HVA points to severe depletion. When DOPAC is high and HVA is low, the problem is rapid intraneuronal degradation. The HVA to DOPAC ratio is what separates the two.
VMA (vanillylmandelic acid)	The primary norepinephrine and epinephrine metabolite, reflecting adrenal catecholamine production.	Low VMA: catecholamine depletion. High VMA: a hyperadrenergic state, often associated with chronic stress and a disrupted cortisol pattern.
HVA to VMA ratio	The ratio of dopamine metabolism to adrenal catecholamine metabolism.	A low ratio indicates dopamine deficiency with intact or elevated adrenal output. This is a common pattern in ADHD presentations: the adrenal system is running hot to compensate for inadequate central dopaminergic function. It is also why so many people describe running on stress.

Serotonin metabolites and the kynurenine pathway

Marker	What it reflects	Interpretation
5-HIAA (5-hydroxyindoleacetic acid)	The primary serotonin metabolite, reflecting production and breakdown rate.	Low 5-HIAA: serotonin depletion. Common associations: tryptophan diversion into the kynurenine pathway from inflammation, gut dysbiosis impairing precursor availability, and shortfalls in B6, magnesium, zinc, or iron. High: rapid turnover, often stress-related.
Kynurenic Acid (KYNA)	Produced when tryptophan is diverted away from serotonin synthesis through the kynurenine pathway, which is activated by inflammatory signals, particularly IDO enzyme induction by inflammatory cytokines.	High KYNA: active inflammatory tryptophan diversion. This is the proposed mechanistic explanation for the depression and anxiety that so often accompany ADHD: the same inflammatory drivers that impair dopamine through the Clostridia pathway simultaneously deplete serotonin through the kynurenine pathway. The supporting nutrients are B2, B6, and B3, cofactors that influence the balance between kynurenic and quinolinic acid.
Quinolinic Acid	An NMDA receptor agonist and excitotoxic kynurenine pathway metabolite, produced when microglia are activated by inflammatory signals.	High quinolinic acid: active microglial activation and neuroinflammation, producing excitotoxicity and oxidative stress.
Quinolinic Acid to 5-HIAA ratio	Compares the excitotoxic arm of tryptophan metabolism to the serotonergic arm.	A high ratio means more tryptophan is going toward neuroinflammatory excitotoxicity than toward serotonin. I read this as a severity marker: the higher the ratio, the more the inflammatory work takes precedence over everything else.

Do not start neurotransmitter precursors until the upstream markers are addressed

This is the single most common sequencing mistake I see, and it is why so many people conclude that amino acid therapy does not work for them.

- **If HPHPA is elevated**, L-tyrosine will produce incomplete results until the Clostridia overgrowth is addressed, because HPHPA interferes with the dopamine synthesis pathway downstream of where tyrosine enters.
- **If fungal markers are elevated**, 5-HTP will be partially metabolized by Candida before it reaches the brain. Address the fungal overgrowth first.

- **If kynurenic acid is high**, adding tryptophan precursors while IDO is driving the kynurenine pathway will send that additional tryptophan toward kynurenic and quinolinic acid, not toward serotonin. The inflammatory driver has to come down first.

Section 13: Energy metabolism and mitochondrial markers

The brain is the most energy-demanding organ in the body, consuming roughly 20 percent of total energy production while making up about 2 percent of body weight. Focus, working memory, processing speed, and emotional regulation all depend on adequate mitochondrial ATP production. When that production is impaired by toxin burden, nutrient shortfall, or fungal metabolites, cognitive performance degrades in a characteristic pattern: inconsistent focus, fatigue that worsens with cognitive effort, and an inability to sustain mental performance through the day. If that description lands, this section is worth your attention.

Krebs cycle metabolites

The Krebs cycle produces the electron carriers that drive the electron transport chain and ATP synthesis. These organic acids give a functional read-out of mitochondrial activity, and abnormal patterns indicate where in the cycle the impairment sits.

Markers	What they reflect	Pattern to look for
Citric Acid, Cis-Aconitic Acid, Isocitric Acid, 2-Oxoglutaric Acid	The early Krebs cycle intermediates.	Elevated citric acid with low downstream metabolites suggests a blockage at the aconitase step, which can be caused by tricarballic acid, the Fusarium metabolite, or by iron-sulfur cluster dysfunction. Elevated citric with low 2-oxoglutaric points there specifically. Check tricarballic acid and consider heavy metal interference. Low citric acid indicates inadequate oxaloacetate input or general cycle underactivity.
Succinic Acid, Fumaric Acid, Malic Acid	The downstream Krebs cycle metabolites. Abnormal patterns indicate specific enzyme dysfunction in the succinyl-CoA to oxaloacetate portion of the cycle.	Elevated succinic with low fumaric suggests succinate dehydrogenase impairment. That enzyme requires FAD, which requires riboflavin. Assess B2 status.

Ketone and fatty acid oxidation markers

Markers	What they reflect	Pattern to look for
Adipic Acid, Suberic Acid, Sebacic Acid, Ethylmalonic Acid	Dicarboxylic acids reflecting fatty acid beta-oxidation. Elevation indicates impaired fatty acid oxidation in the mitochondria.	Elevated suberic and sebacic in particular: impaired beta-oxidation, with carnitine deficiency the most common cause. Elevated ethylmalonic alongside adipic and suberic: electron transport complex impairment. Evaluate riboflavin and carnitine.
3-Methylglutaric Acid, 3-Methylglutaconic Acid, 3-Hydroxyglutaric Acid	Specific mitochondrial markers reflecting HMG-CoA pathway dysfunction.	Elevated 3-methylglutaric and 3-methylglutaconic: impaired HMG-CoA metabolism, which can reflect CoQ10 deficiency or specific complex impairment.
3-Hydroxy-3-Methylglutaric Acid (HMG)	HMG-CoA is the substrate for cholesterol synthesis and ketone body production. Elevated urinary HMG reflects HMG-CoA lyase dysfunction.	Address with riboflavin, carnitine, and CoQ10 support.

Glycolysis markers

Lactic acid and pyruvic acid are glycolysis endpoints. Elevation indicates impaired pyruvate entry into the Krebs cycle, meaning impaired pyruvate dehydrogenase activity. Thiamine is the primary cofactor for that enzyme, so elevated lactic acid points at B1 status, whether from frank deficiency or functional deficiency.

Tier 1 mitochondrial support

- **CoQ10, 200 to 400 mg daily with fat.** An electron transport chain component and the most consistently useful over-the-counter mitochondrial support.
- **Riboflavin (B2), 100 mg daily.** Required for FAD-dependent mitochondrial enzymes and fatty acid oxidation.
- **Benfotiamine (B1), 150 mg twice daily for 4 to 6 weeks.** Benfotiamine is the fat-soluble thiamine form I use. Thiamine is required for pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase.
- **L-carnitine or acetyl-L-carnitine, 1000 to 2000 mg daily.** Transports fatty acids into the mitochondria for beta-oxidation. The acetylated form additionally supports acetylcholine biosynthesis, which is why I lean toward it in cognitive presentations.
- **Magnesium bisglycinate, 300 to 400 mg before bed.** A required cofactor in a very large number of mitochondrial enzyme reactions.

Tier 2 mitochondrial support

Liposomal CoQ10, per label direction, where absorption is a concern, since liposomal delivery outperforms standard capsules. **NMN**, per label direction, as an NAD+ precursor for mitochondrial regeneration.

Section 14: Detoxification and B vitamin markers

Detoxification and oxidative stress

Marker	What it reflects	What I do
Pyroglutamic Acid	Produced during glutathione synthesis in the gamma-glutamyl cycle. Elevation indicates the body is attempting to produce glutathione faster than cysteine can be supplied, so pyroglutamic acid accumulates. A direct marker of glutathione demand outstripping supply.	High: active glutathione depletion, meaning the brain's primary intracellular antioxidant is running short. N-acetyl cysteine as the cysteine precursor, and liposomal glutathione for direct repletion.
2-Hydroxybutyric Acid	Produced during cysteine and glutathione synthesis from homocysteine. Elevation indicates increased catabolic demand on the transsulfuration pathway.	High: oxidative stress driving glutathione demand. Assess methylation status, including methylmalonic acid below for B12. Address the upstream oxidative burden rather than chasing the marker.
N-Acetylcysteine Acid	The urinary form of N-acetyl cysteine.	Low: depleted cysteine availability for glutathione synthesis.
Mandelic Acid	Produced from styrene, a chemical found in plastics, building materials, and food packaging.	High: styrene exposure. Assess environmental sources including polystyrene food containers, synthetic carpet, and outgassing building materials. This is one where finding the source beats supplementing around it. While the source hunt happens, Tox-Chord (Energetix), per label direction, is the homeopathic chemical-exposure formula I use for this finding, with the same honest label as the drainage remedies.
Orotic Acid	A marker of ammonia excess from urea cycle dysfunction, or of folate deficiency. Elevation indicates ammonia accumulation, which is directly neurotoxic.	High: ammonia burden with neurotoxic implications. Liver support becomes a priority, and Hepata-Chord (Energetix), per label direction, is the homeopathic liver support formula I use for it, with the same honest label as the drainage remedies. Assess B12 and folate status. The B12 combination from the drainage phase is directly supportive here.

B vitamin and nutritional markers

Marker	What it reflects	What I do
Methylmalonic Acid (B12 marker)	Accumulates when the B12-dependent enzyme methylmalonyl-CoA mutase cannot function. A functional indicator of B12 deficiency even when serum B12 looks normal. B12 is required for myelin synthesis, DNA methylation, and homocysteine clearance.	High: functional B12 deficiency. Serum B12 can read normal while functional deficiency impairs neurological function, which is exactly why I order this marker. Sublingual or liposomal methylated B12 at 2000 mcg, run alongside folate, B6, and TMG, is my preference over oral, particularly with compromised gut function.
Pyridoxic Acid (B6 marker)	The primary urinary B6 metabolite. B6 is an essential cofactor for the decarboxylase enzymes that produce dopamine, serotonin, GABA, and norepinephrine from their amino acid precursors.	Low: B6 deficiency impairing neurotransmitter synthesis across every catecholamine and serotonin pathway at once. This is one of the most consistent reasons I find for treatment-resistant low HVA and low 5-HIAA. Address B6 before adding any neurotransmitter precursor.
Glutaric Acid (B2 marker)	Accumulates when riboflavin-dependent enzyme function is impaired, particularly glutaryl-CoA dehydrogenase.	High: functional B2 deficiency impairing mitochondrial fatty acid oxidation and the Krebs cycle at FAD-dependent steps.
Methylcitric Acid (biotin marker)	Accumulates when propionyl-CoA carboxylase is impaired. That enzyme requires biotin.	High: functional biotin deficiency. Biotin is required for multiple carboxylase enzymes essential to glucose, amino acid, and fatty acid metabolism.
Oxalic Acid, Glyceric Acid, Glycolic Acid	Oxalate pathway markers. Elevated oxalates can result from fungal overgrowth, since Aspergillus produces oxalic acid, from B6 deficiency reducing enzymatic clearance, and from high dietary oxalate intake. Oxalate crystals deposit in tissue, including brain tissue.	High oxalates: fungal overgrowth is the most common cause in this population, so address Candida and Aspergillus first rather than restricting oxalate foods indefinitely. Support clearance with B6 and calcium taken with meals to bind dietary oxalate.

Section 15: Reading your results as a pattern

Individual elevations matter, but the most actionable information comes from the pattern across both panels. These combinations are the ones I encounter most often in attention and fog presentations, and each points toward a specific primary driver.

Pattern	What it suggests	Where I start
Anti-S100B and Anti-Microglia elevated, arabinose high, fungal OAT markers elevated	Candida or Aspergillus overgrowth driving gut and barrier permeability, with fungal metabolites crossing into the brain and provoking inflammatory antibody production.	Start with antifungal work (pau d'arco, oregano oil within the daily cap, and <i>Saccharomyces boulardii</i>), with drainage established before any of it begins.
Anti-DR1 and Anti-DR2 elevated, Anti-Streptococcal A elevated, HPHA high	Consistent with the PANDAS and PANS mechanism, where streptococcal molecular mimicry drives dopamine receptor antibodies, with concurrent Clostridia overgrowth suppressing dopamine synthesis directly.	Address the streptococcal burden with berberine, and the Clostridia with gut barrier repair and probiotic restoration. Both run concurrently, after drainage.
Anti-EBV or Anti-HHV-6 elevated, barrier markers elevated, quinolinic acid high with low 5-HIAA	Reactivated herpes virus driving barrier disruption and microglial activation, with the resulting inflammation activating the kynurenine pathway and pulling tryptophan away from serotonin toward excitotoxic quinolinic acid.	Viral and thymic immune support. Bioavailable curcumin for the inflammatory arm. Address the cortisol pattern, since cortisol suppresses antiviral immunity.
Multiple demyelination markers elevated, Anti-MBP or Anti-MOG positive, low 5-HIAA	Active demyelinating autoimmunity, most likely gluten molecular mimicry, with serotonin depletion partly downstream of the gut permeability driving the autoimmune response.	Strict gluten elimination, and this one is not negotiable. Wheat Zoomer to characterize it. Gut barrier repair with L-glutamine-based support.
Tartaric acid, 3-oxoglutaric acid, and furanic compounds elevated, with Anti-S100B or Anti-Microglia elevated	Environmental mold exposure driving Aspergillus colonization, with the OAT markers indicating the gut colonization that would explain the barrier disruption and neuroinflammation.	Order the Mycotoxins panel. Investigate the environment, including ERMI dust testing of home and workplace. Mold work becomes the priority, with binder support and correct binder timing.
Low HVA and low 5-HIAA, Neural Zoomer negative, HPHA or arabinose elevated	Neurotransmitter depletion driven primarily by gut microbial metabolites rather than by autoimmune receptor attack. This is the most straightforward version of the picture.	Antifungal or anti-Clostridia work per the specific OAT findings, plus gut barrier repair and drainage. Neurotransmitter precursors become appropriate once the gut dysbiosis is addressed.

Section 16: Your next steps

Your Neural Zoomer Plus and OAT results give you the upstream picture that most people with attention and fog complaints never get offered. Immune reactivity data and metabolic data read together point at the specific candidates, whether fungal overgrowth, viral reactivation, Clostridia metabolites, mold burden, gut permeability, or nutrient depletion.

The most important thing I can tell you is not to address every finding at once. **Get the order right.** Drainage comes first. Antimicrobial, antifungal, and antiviral work comes only after drainage is established.

Neurotransmitter precursor support comes only after the microbial drivers of neurotransmitter depletion are addressed. Shortcutting that order produces incomplete results and, often enough, a stretch of feeling worse for no gain.

And hold all of it to an endpoint. Keep an honest log of the two or three things you most want to change, whether that is sustained focus, afternoon crash, sleep, or emotional regulation. Compare it against your own baseline, not against anyone else's numbers. Retest when it makes sense to. If the markers move and your log does not, we were chasing the wrong thing, and I would rather find that out in eight weeks than in two years.

On the formulas named in this guide, and the few that are not

You will notice this guide names generic actives, meaning curcumin, baicalin, pau d'arco, glutathione, and so on, alongside a small set of named Energetix formulas, and that in a few places it describes a clinical role without naming a product at all.

There is a reason for that mix. The protocol this guide grew out of was built around a practitioner-exclusive product line that I have moved away from, on practical grounds rather than clinical ones: it could not be ordered directly by patients, which meant anyone outside my immediate area could not reliably get it. Everything I recommend now runs through my Fullscript dispensary so you can obtain it wherever you live, and each Energetix formula named in this guide is carried there. For a few roles in the original protocol, the astrocyte and glymphatic remedy, the thymic and colostrum-based immune pair, and the gut immune formula among them, I have not finished selecting the replacement I want to stand behind, and I am not going to put a substitute in your hands just to fill a line in a table. Where that applies, I have said so. Bring it up in the Discovery Session and I will tell you exactly what I am using now.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

The Root Discovery Session is free. It is a clinical conversation that walks through your specific results, identifies the highest-priority drivers, and maps the right sequence for your case.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we

would work together from there.

If you would rather start on your own, begin with two weeks of honest tracking against the drainage foundation, and bring the log with you whenever you are ready.

Free tracker: rootcauseunlocked.com/tracker

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Educational content only. Not medical advice. Not a diagnosis, and this guide does not establish a patient-provider relationship. Never stop, skip, or change a prescription based on anything in here.

Read this part first

This guide is for general education. It does not diagnose anything, and reading it does not make you my patient.

Diet does not cause ADHD, and there is no universal ADHD diet that cures it. Anyone selling you one is selling you something. What follows is terrain work, which is a different and more modest claim.

Nothing here replaces an individualized dietitian consult, a physician-directed food allergy workup, or your prescriber's guidance on medication. Talk with me before you make a large or restrictive dietary change, and especially before you make one for a child.

If you are in crisis right now, call or text **988** (Suicide and Crisis Lifeline, available 24 hours a day). Food can wait. That call cannot.

Why diet is part of the terrain

Real time spent on nutrition is still worth it, for two reasons that have nothing to do with curing anything.

The first is that the brain depends on adequate energy, amino acids, essential fatty acids, vitamins, and minerals, and blood sugar stability is one of the terrain factors that most directly amplifies or calms an ADHD day. The second is that ADHD itself makes consistent eating harder. Forgetting to eat, low appetite from medication, sensory preferences, impulsive snacking, and the difficulty of stopping an activity long enough to make food are all part of the picture, and they compound each other.

So there are two levers that matter most, and this guide is built around them. Keep blood sugar and protein steady through the day, and support the gut-brain connection, since certain gut bacterial and yeast patterns can interfere with dopamine processing in ways a symptom checklist alone will never surface.

I want to be plain about where national guidance sits, because it shapes how I use this. NICE guideline NG87 advises balanced nutrition and regular exercise, and it does not recommend routine elimination of artificial colors or additives, routine fatty-acid supplementation as a treatment, or unsupervised few-food diets for every child with ADHD. This guide follows that same caution. Individualized, testing-informed changes, not a blanket restriction applied to everyone.

Part one: blood sugar and protein timing

Build meals around four anchors

- **Protein.** Poultry, fish, lean meat, Greek yogurt, tofu, beans, lentils, eggs, or another source you tolerate.
- **Fiber-rich carbohydrate.** Oats, potatoes, brown rice, quinoa, whole-grain bread, beans, fruit, or vegetables.
- **Healthy fat.** Olive oil, avocado, nuts, seeds, or fatty fish.
- **Color and micronutrients.** Fruits and vegetables in forms the person will actually eat.

Protein deserves the emphasis, and I want to state the claim honestly rather than overselling it. A protein-containing meal reduces rapid hunger, supports overall nutrient intake, and makes medication-related appetite changes easier to manage. That is the defensible version. Protein does not treat ADHD directly.

Combining carbohydrate with protein, fat, and fiber produces a steadier meal than refined carbohydrate on its own. Highly restrictive low-carbohydrate plans are not necessary for most people with ADHD, and the restriction usually costs more in adherence than it returns in steadiness.

Medication and appetite

Stimulant medication can reduce appetite, especially in the middle of the day. Children taking stimulants need growth monitoring. Adults should watch for unintended weight loss, under-fueling, headaches, irritability, and evening rebound eating.

A substantial breakfast before the medication, portable calorie-dense snacks, and a nourishing evening meal all help. If you suspect a medication is driving an appetite problem, that belongs in a conversation with the prescriber. It does not belong in an eating pattern that quietly gets more restrictive to compensate.

A simple day of eating

- **Breakfast.** Greek yogurt, berries, oats, and walnuts, or eggs with whole-grain toast and fruit.
- **Lunch.** Rotisserie chicken or chickpeas, microwave rice, bagged salad, and an olive-oil dressing.
- **Snack.** Apple with peanut butter, cheese with whole-grain crackers, or a protein smoothie.
- **Dinner.** Salmon, chicken, tofu, or beans with potatoes or rice and a frozen vegetable.
- **Emergency meal.** A low-sugar protein drink, fruit, nuts, and a whole-grain snack.

If you are someone who forgets to eat, lean on external cues rather than willpower. Pair meals with fixed events that already happen. Keep visible snacks where the decisions get made. Use recurring alarms. Keep a short list of zero-preparation foods on hand. Convenience is a therapeutic feature here, not a shortcut you should feel bad about.

Part two: supporting the gut-brain connection

Laboratory work indicates that specific gut bacterial and yeast overgrowths produce byproducts capable of interfering with dopamine processing. That is a proposed mechanism rather than a settled one, and it is not a proven treatment pathway for ADHD.

Here is what I do with it anyway, because the gap does not leave me with nothing. Rather than guessing which food or supplement to try next, I test. An Organic Acids Test, alongside a neurological antibody panel, tells me where to look. Those are functional markers, so they are suggestive rather than diagnostic, and I read them as a direction to investigate rather than as a verdict. Once results are back, specific findings get addressed individually, as part of a plan built for one person rather than for a category.

The food-based supports below are the general, everyday version of that same gut-brain principle, and they are safe to start on without any testing at all.

Everyday gut support

- Reduce added sugar and highly processed refined carbohydrate, which act as a growth substrate for some gut organisms.
- Build in adequate fiber and a diverse range of plant foods, for a more resilient microbiome.
- Introduce fermented foods in small amounts once you know your own tolerance. Go slowly if you are not sure how you respond.
- Prioritize wild-caught fatty fish, meaning salmon, sardines, and mackerel, a few times weekly, for the same omega-3s covered in the ADHD Supplement Guide.

Nutrient-dense foods worth building in

The brain needs adequate vitamins and minerals to run, and food is the least complicated place to get them. These are the ones I like to see covered on the plate before anything gets added in a capsule.

For **magnesium**, look to pumpkin seeds, leafy greens, avocado, and dark chocolate. For **zinc**, pumpkin seeds again, along with grass-fed beef, chickpeas, and oysters if you tolerate them. For **B vitamins**, specifically B1, B2, and B6, look to eggs, poultry, leafy greens, whole grains, and legumes. For **omega-3s**, wild-caught salmon, sardines, mackerel, walnuts, and ground flaxseed.

Part three: a structured method for suspected food sensitivities

When a specific food seems to connect repeatedly to behavior or to physical symptoms, the safer next step is a structured diary and a dietitian-guided evaluation. Not guesswork, and not a blanket elimination diet. Use this four-step method instead of removing foods on suspicion alone.

Step one, track before you change anything. Keep a simple daily food-and-symptom log for 2 to 3 weeks. What was eaten, when, and how the day went for focus, mood, and behavior.

Step two, look for a pattern across multiple exposures. One rough afternoon after a food is not evidence. The same reaction on several separate occasions is worth pursuing.

Step three, remove one food, then put it back. If a specific food repeatedly correlates with a symptom, take that one food out for 2 to 3 weeks, then reintroduce it once and observe the response. Do this with a dietitian's guidance or mine, rather than on your own.

Step four, check the pattern against testing where testing applies. If gluten is the suspected pattern, for example, a Wheat Zoomer can clarify gluten cross-reactivity. A result like that is consistent with a pattern rather than proof of one, and it is still a great deal better than committing to permanent broad restriction on suspicion alone.

This is not a universal elimination diet, and I do not want it read as one. National guidance does not support routine elimination of artificial colors, additives, or unsupervised few-food diets for every child with ADHD. The method above exists for a specific, repeatedly observed pattern in one individual.

If you are watching for a hyperactivity or behavior pattern in particular, artificial food dyes (Red 40, Yellow 5, Yellow 6) and sodium benzoate are reasonable first suspects to track. These carry required warning labels in

the EU related to hyperactivity in some children. Track them with the same diary method above rather than pulling every processed food at once, which tells you nothing except that a lot changed.

Part four: a simple daily framework

If you remember nothing else from this guide, remember this shape.

- **Protein at every meal.** Poultry, fish, lean meat, Greek yogurt, tofu, beans, lentils, eggs.
- **Fiber-rich carbohydrate at every meal.** Oats, potatoes, brown rice, quinoa, whole-grain bread, beans, fruit.
- **Healthy fat at every meal.** Olive oil, avocado, nuts, seeds, fatty fish.
- **Color and micronutrients daily, varied.** Fruits and vegetables in forms you will actually eat.
- **Hydration daily.** As a working gauge, half your body weight in ounces of filtered water.

What this guide deliberately does not do

It does not replace an individualized dietitian consult, and it does not diagnose a food allergy. A food allergy diagnosis requires physician-directed testing, and nothing in a symptom diary substitutes for it.

It does not recommend a restrictive elimination protocol, such as an autoimmune protocol or a few-food diet, outside of clinical supervision. Any elimination that goes beyond the structured method in part three should be guided by me or by a registered dietitian, and it should not be self-directed long term.

It does not go into specific gut-organism findings, meaning Clostridia, Candida, and their related metabolites, or the targeted dietary and supplement response to each. Those are individual calls that depend on what your functional testing actually shows, and they are worth an appointment rather than a paragraph. Bring your results and we will work through them together.

A note on safety

Seek urgent medical care for suicidal thoughts, psychosis, severe agitation, chest pain, fainting, a dangerous medication reaction, or any behavior that creates immediate risk. In the United States, call or text 988 for crisis support, and call 911 for an immediate emergency.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your history, and your medication list.

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Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

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If you would rather start on your own:

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Read the companion documents alongside this one. **The ADHD Supplement Guide** covers over-the-counter level support and the omega-3 evidence in detail.

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Educational content only. Not medical advice. Not a diagnosis. Never stop, skip, or change a prescribed medication based on this guide. In crisis, call or text 988 now.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose ADHD, it does not replace a comprehensive evaluation, and reading it does not make you my patient.

ADHD is a real neurodevelopmental condition, and nothing here is meant to talk you out of medication or to position a supplement as a replacement for one. Never start, stop, or change ADHD medication, or any other prescription, based on this guide. That conversation belongs with the physician managing that part of your care.

Talk with your prescriber, and with me, before starting anything here, especially for a child, during pregnancy, or alongside ADHD medication. Supplements can cause adverse effects and can interact with medicines.

Seek urgent medical care for suicidal thoughts, psychosis, severe agitation, chest pain, fainting, a dangerous medication reaction, or behavior that creates immediate risk. In the United States, call or text **988** for crisis support and call **911** for an immediate emergency. No supplement has any role in a crisis.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for third party tested products and age appropriate dosing.

How this guide works

Focus, follow through, and emotional regulation are shaped by more than genetics. They are shaped by gut function, inflammation, nutrient status, blood sugar stability, sleep, and how well the brain's detox and energy systems are keeping up. That is the terrain, and it is where I start. Addressing terrain does not replace medical care. It gives medication, therapy, and every other tool a better chance to work.

The strongest reason to use any nutrient in my practice is a documented deficiency, a specific clinical need, or a carefully reviewed trial with realistic expectations. Never a hunch.

Most people need one or two things from the list below, not all of them. Individualize rather than stacking everything at once. Basic bloodwork with your provider, or the broader Neural Zoomer Plus and Organic Acids Test panel I run and interpret myself, tells you which of these actually apply to you. Those functional panels are suggestive rather than diagnostic, which is exactly how I use them: as a map of where to look, read alongside your history and exam rather than in place of them.

Why the evidence here looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money exists only when a patent can pay it back. Nobody can patent iron, magnesium, or fish oil. So the evidence hierarchy systematically over documents patentable drugs and under documents nearly everything else, which is worth knowing before you read a null result as a verdict.

The honesty runs in both directions, though, and I will not use funding bias as a license to claim more than I can. Where a real trial ran and the answer was no, this guide says so plainly and then tells you what I do instead. Small nutrient trials carry their own bias toward positive findings. One rule, both directions.

How to read this guide

Give the foundations two weeks before adding anything targeted. Drainage first, without exception. This is the most commonly skipped step and the one that makes the rest work.

One change at a time. If you start four things in a week you will never know which one earned its place, and you will not know which one caused a reaction either.

Eight to twelve weeks is a fair trial for most of what follows, then reassess honestly against how you actually functioned at baseline, not against how you hoped you would feel.

Track it. A short daily note on focus, follow through, sleep, and energy is worth more to me at a visit than a bottle count.

This is a starting point, not the finish line. Full personalization is guided by testing and by your own response, not by a general list.

Foundations first

These come before any product, and they cost almost nothing.

Support	How to use it
Dry brushing	5 to 10 minutes before showering, strokes toward the heart
Castor oil packs	Over the liver, 3 to 4 times weekly
Hydration	Half your body weight in ounces of filtered water daily
Light rebounding	10 minutes daily

On drainage remedies. Where a patient needs more than the four supports above, the drainage line I use is Energetix, chosen because it is carried on Fullscript and you can order it yourself rather than waiting on me. The standard mapping is Lymph-Tone I for acute presentations, Lymph-Tone II for matrix and most chronic cases, Lymph-Tone III for deep or longstanding cases, and ReHydration for electrolytes and cellular hydration. Take these per label direction. I do not carry a dose from one brand to another, and I have not published my own dosing for this line yet, so until I do, the label is the honest answer.

These are homeopathic remedies, and I will be straight with you about what that means: the research base does not support homeopathy by the standard I apply to everything else in this guide. I use these anyway, because in my clinical experience patients tolerate the drainage phase better and move through it more

smoothly with them than without, and because the cost and the risk are both low. That is clinical observation, not trial evidence, and I am labeling it as such. Which formula fits you is individualized, and the selection is worth a conversation rather than a guess. Choose one. Never stack them.

Correcting a documented deficiency

These are the nutrients most likely to actually be low in someone with ADHD or brain fog, and the ones where correcting a real, documented deficiency, rather than simply taking a supplement, tends to produce the clearest benefit.

Omega 3 fatty acids, EPA forward

What it is and why. Omega 3s, particularly EPA, are the most extensively studied nutritional supplement for ADHD. A 2023 meta-analysis of 22 randomized controlled trials found no statistically significant overall improvement in core ADHD symptoms compared with placebo, though trials lasting at least four months showed a modest subgroup benefit (reference 1). Omega 3s also support the phospholipid membranes that make up the blood brain barrier and help calm general neuroinflammation.

The gap, and what I do with it. That is a null result on core symptoms, and I am not going to dress it up. What it did not test away is the membrane and barrier role, which is why omega 3 stays one of my general terrain supports regardless of formal testing, alongside your main ADHD care rather than instead of it. If you are running it specifically as a symptom adjunct, the four month subgroup signal is the reason to give it a real trial length rather than three weeks.

Who it fits. A reasonable option for anyone with low fish intake who wants a well tolerated adjunct. It is not a substitute for medication or behavioral treatment.

Typical over the counter dose and form. For general nutritional and blood brain barrier support, 2,000 to 3,000 mg daily of combined EPA and DHA, triglyceride form. When used specifically as an ADHD symptom adjunct, the doses used in trials are lower: commonly 500 to 1,000 mg a day in children and adolescents and 1,000 to 2,000 mg daily in adults, typically trialed for 8 to 12 weeks before reassessing.

Safety and interactions. GI upset, fish allergy, and bruising or bleeding risk at higher doses or alongside blood thinners or antiplatelet medication. Pause before scheduled surgery. Check the label for actual EPA and DHA content, not total fish oil weight.

Iron, only with a confirmed deficiency

What it is and why. Iron is required for oxygen transport and plays a role in dopamine related brain chemistry. Low iron or low ferritin is a well documented, common condition that can produce or worsen attention problems on its own, including fatigue, restless legs, and poor concentration.

Who it fits and testing. Only for a confirmed low ferritin or iron deficiency on bloodwork (CBC, ferritin, and iron with TIBC or transferrin saturation). Reasons to test include restricted eating, heavy menstrual bleeding, a vegetarian or vegan diet, fatigue, restless legs, pallor, slow growth, or a prior history of low ferritin. Iron should never be started just because a person has ADHD.

Typical over the counter dose and form. Once a deficiency is confirmed, adult dosing commonly runs 40 to 65 mg elemental iron once daily or every other day. Children need weight based dosing, roughly 2 to 3 mg per kg per day of elemental iron, set by their pediatrician. Bloodwork is typically rechecked in 6 to 12 weeks.

Safety and interactions. Iron is toxic in overdose, and accidental ingestion by a child can be fatal. Keep every iron containing product locked away from children. Iron commonly causes constipation and nausea. Separate iron dosing by 2 to 4 hours from levothyroxine, tetracycline or quinolone antibiotics, calcium, magnesium, and zinc, since they block each other's absorption. Any confirmed deficiency deserves an investigation into its cause, not just a bottle of pills.

Zinc

What it is and why. Zinc is a direct modulator of dopamine metabolism and a cofactor the brain relies on for normal neurotransmitter signaling.

Who it fits and testing. Best fit for a documented deficiency, low dietary intake, malabsorption, or a carefully monitored trial, ideally informed by a micronutrient panel or the Organic Acids Test.

Typical over the counter dose and form. Zinc bisglycinate, 15 to 30 mg with food, generally trialed 8 to 12 weeks in adolescents and adults when indicated. Children usually need a lower range, about 5 to 15 mg per day elemental, based on age and status.

Safety and interactions. Nausea. Chronic higher dosing can cause copper deficiency, so copper should be monitored with prolonged use above routine requirements. Separate from tetracycline or quinolone antibiotics.

Magnesium, glycinate

What it is and why. Magnesium deficiency is common, and magnesium is a cofactor for the calming, inhibitory brain chemistry of the GABA pathway as well as for hundreds of mitochondrial energy production reactions.

The gap, and what I do with it. ADHD specific evidence for magnesium alone is limited, so I do not present it as a targeted ADHD treatment. I use it as a general terrain support, and I use it often, because the presentations it does have evidence for, poor sleep, muscle tension, constipation, and migraine, travel with brain fog constantly and are worth fixing on their own terms.

Who it fits. Low dietary intake, deficiency risk, constipation, migraine, muscle tension, or sleep support.

Typical over the counter dose and form. Magnesium glycinate or threonate, 300 to 400 mg before bed, is the form and timing I use most often. Deficiency correction ranges can run lower, roughly 100 to 300 mg elemental magnesium in the evening in adults, titrated to bowel tolerance. Children run roughly 5 to 10 mg per kg per day elemental.

Safety and interactions. Diarrhea at higher doses. Avoid high doses with significant kidney impairment. Separate from levothyroxine, bisphosphonates, tetracyclines, and quinolones.

Vitamin D3

What it is and why. Vitamin D status affects far more than bone health, including immune and general neurological function. The goal here is correcting a documented insufficiency, not treating ADHD directly with vitamin D.

Who it fits and testing. Limited sun exposure, low dietary intake, malabsorption, obesity, or a documented prior deficiency. Test your level with your provider before and during treatment. Do not guess at a dose without a number.

Typical over the counter dose and form. Maintenance doses commonly run 600 to 1,000 IU a day in children and 1,000 to 2,000 IU daily in adults. Treating a confirmed deficiency uses a higher, guideline based loading dose that your provider sets from your measured level.

Safety and interactions. Excessive dosing risks high blood calcium and kidney injury. Use caution in granulomatous disease, hyperparathyroidism, kidney stones, or with thiazide diuretics. Recheck your level when treating a deficiency.

Broad spectrum micronutrients

What it is and why. Broad vitamin and mineral formulas, more comprehensive than a standard multivitamin, have produced promising but mixed findings. In one randomized trial, clinician rated global improvement was greater with a broad spectrum micronutrient formula than with placebo, while the primary parent rated symptom measure was not significantly different (reference 2). Earlier randomized trials in both adults and children reported benefit on some outcomes but not consistently across every rater (references 3 and 4).

The gap, and what I do with it. Mixed across raters is genuinely mixed, and I say so before anyone spends on it. Where I do reach for this category is the person whose diet has been poor for years and whose panel shows several small insufficiencies rather than one clear hole, because chasing five separate bottles is worse medicine than one studied formula. It is not first line, and it does not replace correcting a specific documented deficiency.

Typical over the counter dose and form. Use a studied commercial formulation and follow its specific label and trial protocol rather than combining several single high dose nutrients on your own.

Safety and interactions. High pill burden. Risk of duplicating vitamin A, D, B6, zinc, selenium, and iron if combined with other products. Review every ingredient against your other supplements and medications before starting.

Blood brain barrier and neuroinflammation support

The blood brain barrier is the gatekeeper between the rest of the body and the brain. When it is compromised, the brain is exposed to more of the body's inflammatory and toxic burden. These are foundational, general nutrition level supports, not a diagnosis specific prescription.

Lion's mane mushroom extract

What it is and why. Supports nerve growth factor production, which plays a role in barrier repair and general neurological maintenance.

Who it fits. General nutritional terrain support, not gated to a specific test. Appropriate for anyone interested in adding a well tolerated cognitive support supplement.

Typical over the counter dose and form. 500 to 1,000 mg daily.

Safety and interactions. Generally well tolerated. Rare GI upset or allergic reaction. Long term safety data are still limited, which is why I use a reputable dual extract product, keep it inside the dose above, and treat it as one piece of a barrier plan rather than a standing lifetime supplement.

Phosphatidylserine and phosphatidylcholine

What it is and why. Both are phospholipids the brain uses to build and repair neuronal cell membranes. Phosphatidylserine supports general neuronal membrane repair. Phosphatidylcholine, a choline rich phospholipid, supports the same membrane biosynthesis and is especially relevant when processing speed and sensory sensitivity complaints are prominent.

Who it fits. General nutritional terrain support, not gated to a specific test. A reasonable adjunct alongside omega 3 for anyone focused on the membrane and barrier category.

Typical over the counter dose and form. Phosphatidylserine 100 mg three times daily, or phosphatidylcholine 1,200 mg daily as an alternative or addition.

Safety and interactions. Generally well tolerated. Occasional mild GI upset. Most commercial products are soy or sunflower derived, so check for allergy.

Dietary supports worth knowing about. Removing seed oils, meaning canola, soybean, corn, and sunflower, and watching your own pattern around artificial food dyes such as Red 40, Yellow 5, and Yellow 6, and around sodium benzoate. Those additives carry required warning labels in the European Union for hyperactivity effects in some children. Gluten elimination is covered separately in **The ADHD Diet Guide**. It is testing informed here, not a blanket recommendation for everyone with ADHD.

Mitochondrial and energy support

The brain consumes a disproportionate share of the body's energy production. When cellular energy production is impaired, the pattern is often inconsistent focus, fatigue that worsens with mental effort, and difficulty sustaining performance through the day.

CoQ10

What it is and why. A component of the cell's electron transport chain, and the most consistent over the counter mitochondrial support available.

Who it fits. General energy terrain support, and particularly worth raising with your provider if you take a statin, since statins can lower CoQ10 levels.

Typical over the counter dose and form. 200 to 400 mg daily, taken with a fat containing meal for absorption.

Safety and interactions. Generally well tolerated. Rare GI upset. May modestly lower blood pressure or affect warfarin dosing, so tell your provider if you take a blood thinner.

Riboflavin, vitamin B2

What it is and why. Required for the FAD dependent mitochondrial enzymes involved in fatty acid oxidation and general cellular energy production.

Who it fits. General energy terrain support. No specific gating test required.

Typical over the counter dose and form. 100 mg daily.

Safety and interactions. Very safe at this dose. Harmless bright yellow urine is common and expected.

Thiamine, vitamin B1, as benfotiamine

What it is and why. Required for pyruvate dehydrogenase, a key step that moves fuel into the mitochondria for energy production. Benfotiamine, a fat soluble form of thiamine, is the form I use.

Who it fits. General energy terrain support. Worth particular attention with a history of poor diet, heavy alcohol use, or bariatric surgery.

Typical over the counter dose and form. Benfotiamine, 150 mg twice daily for 4 to 6 weeks, then reassess.

Safety and interactions. Very safe at this dose, with no major interactions at typical intake.

L-carnitine or acetyl-L-carnitine

What it is and why. Transports fatty acids into the mitochondria for energy production. The acetylated form also supports acetylcholine production, which is relevant when memory and processing speed are the primary complaints.

Who it fits. General energy terrain support, most relevant when fatigue and inconsistent mental stamina are prominent. When memory, processing speed, and word finding lead the picture, the acetylated form is my first choice in this category.

Typical over the counter dose and form. 1,000 to 2,000 mg daily.

Safety and interactions. GI upset or a fishy body odor at higher doses. Talk with your provider if you take a blood thinner or have a thyroid condition.

Magnesium belongs here too. Magnesium glycinate, 300 to 400 mg before bed, is foundational for energy as well as sleep, since magnesium is a required cofactor in hundreds of mitochondrial enzyme reactions. See the magnesium entry above for the full safety and interaction picture, and do not run two magnesium products at once.

Oxidative stress support

N-acetyl cysteine, NAC

What it is and why. Supplies the raw material, cysteine, for the brain's primary antioxidant, glutathione. It is one of the markers I look at directly on functional testing.

The gap, and what I do with it. NAC's efficacy specifically for core ADHD symptoms is not established, and I do not use it as an ADHD intervention. I use it where the testing points at oxidative stress or strained detox capacity, which is a separate, defined indication that stands on its own whether or not it changes an attention score.

Typical over the counter dose and form. 600 to 1,200 mg once or twice daily.

Safety and interactions. GI upset. Use caution with nitroglycerin, where it adds headache and low blood pressure risk, and in asthma.

Gut support

Probiotic, Lactobacillus and Bifidobacterium blends

What it is and why. A meaningful share of the body's serotonin is produced in the gut, and specific bacterial imbalances can produce byproducts that interfere with dopamine metabolism. Supporting a healthy gut bacterial population is a foundational, low risk terrain step.

Who it fits and testing. A general gut support option for most patients. More specifically indicated when gut focused functional testing such as the Organic Acids Test shows patterns consistent with a bacterial or yeast imbalance.

Typical over the counter dose and form. Follow the product label. Multi strain Lactobacillus and Bifidobacterium formulas are the type most commonly used for this purpose, since studied colony count ranges vary widely by strain and by product. I would rather send you to the label than invent a number the research does not support.

Safety and interactions. Generally well tolerated. Mild bloating or gas is common when starting. Use caution and check with your prescriber if you have a central line or are significantly immunocompromised.

Immune support

These are relevant to some patients' testing patterns around viral reactivation. Zinc and vitamin D also belong in this conversation and appear above.

Vitamin C

What it is and why. A general immune function support, relevant to patients whose testing or history points toward viral reactivation patterns.

Who it fits. General immune support, most useful in the context of recurrent viral symptoms or a provider identified immune concern.

Typical over the counter dose and form. 1,000 mg two to three times daily.

Safety and interactions. GI upset or loose stools at higher doses. Sustained high intake can raise kidney stone risk in susceptible individuals.

Lysine

What it is and why. May help manage herpes family viral activity by competing with the amino acid arginine that these viruses need to replicate.

Who it fits. Most relevant for patients with a history of cold sores or other herpes family viral reactivation patterns.

Typical over the counter dose and form. 1,000 mg twice daily on an empty stomach.

Safety and interactions. Generally well tolerated. Rare GI upset. Discuss with your provider if pregnant.

Sleep support

Melatonin

What it is and why. A sleep timing tool, not a core ADHD treatment. It can help some children or adults with delayed sleep onset, particularly combined with a consistent bedtime routine.

Who it fits. Delayed sleep onset, after a behavioral sleep plan is already in place. Children should use melatonin only with pediatric guidance, especially for long term use.

Typical over the counter dose and form. Children and adolescents typically start at 0.5 to 1 mg, 30 to 90 minutes before the target bedtime, commonly landing at 1 to 3 mg, and going higher only with specialist guidance. Adults commonly use 1 to 5 mg. Because timing matters more than quantity here, more is not more effective.

Safety and interactions. Morning sedation, vivid dreams, and variable product quality. If you or your child take guanfacine, clonidine, or a sedating antihistamine, talk with the prescriber first, since melatonin can add to sedation.

Approach these with caution, every time

- High dose single nutrients started without testing or a clear indication.
- St. John's wort, because of major medication interactions and inadequate ADHD evidence.
- "Dopamine boosters," mucuna or levodopa products, and proprietary nootropic blends with undisclosed doses.
- Unmonitored high dose vitamin B6, or megadose methylfolate based only on an MTHFR gene result.
- Caffeine as self medication, especially in children or alongside stimulant medication.
- Iron containing products kept where children can reach them. Iron toxicity is a real risk and accidental ingestion can be fatal in children. Start iron only with a confirmed deficiency and medical guidance. See the iron entry above.
- Any product claiming to cure ADHD, detox the brain, or replace prescribed treatment. Be skeptical of anyone who makes those claims, including me.

What this guide covers, and what it does not

This is the foundational, over the counter tier. Everything in it is something you can act on yourself, with your provider's guidance, without a lab result in hand. That is deliberate. This part of the plan does not need me standing next to it.

What it does not include is the personalized sequencing, the antifungal and antimicrobial support, and the practitioner grade products I use once functional testing identifies your specific upstream drivers, gut dysbiosis, mold burden, and viral reactivation among them. That level of care is built individually, from your own lab results, and it is not something I would put in a general list for a reader I have never examined.

I have also left out gray market "smart drugs," high dose single amino acids used for neurotransmitter loading, and any blend whose mechanism is described in marketing language rather than chemistry. A guide reaches you without a history, a medication list, an exam, or any way for me to change course next week, so the bar for what goes in it is higher than the bar for a conversation.

Safety and medication note

Never start, stop, or change ADHD medication, or any other prescription, based on this guide. Coordinate all supplement use with your prescriber, especially for children. Supplements can cause adverse effects and can interact with medicines. Choose third party tested products and age appropriate dosing.

Seek urgent medical care for suicidal thoughts, psychosis, severe agitation, chest pain, fainting, a dangerous medication reaction, or behavior that creates immediate risk. In the United States, call or text 988 for crisis support and call 911 for an immediate emergency.

Where to find quality versions of these

Quality varies, and this category is worse than most for it. Look for independent third party testing, meaning NSF, USP Verified, or an equivalent seal, whichever source you use. Check the dose and the form against the entries above rather than against the front of the bottle.

I hold no physical inventory. I keep a screened dispensary through Fullscript, which ships directly to you:

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. Nothing in this guide was included, excluded, ranked, or dosed based on what it earns. Worth noticing: the four foundations at the top of this guide, and the sleep and diet changes behind them, earn me nothing at all. That is still where I want you to start.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your history, your labs, and your full medication list, and it sequences interventions rather than stacking them.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, track first. Four weeks of daily notes on focus, follow through, sleep, and energy is what turns “I think it helped” into a decision:

Free symptom tracker: rootcauseunlocked.com/tracker

Read the companion documents alongside this one. **Understanding Your ADHD and Brain Fog Labs** covers what to test before you supplement, and **The ADHD Diet Guide** covers the food changes, including the gluten question.

References

Verification detail for these entries is documented in `ADHD-Citation-Ledger.md`.

1. Liu TH, Wu JY, Huang PY, Lai CC, Chang JPC, Lin CH, Su KP. Omega-3 Polyunsaturated Fatty Acids for Core Symptoms of Attention-Deficit/Hyperactivity Disorder: A Meta-Analysis of Randomized Controlled Trials. *J Clin Psychiatry*. 2023;84(5):22r14772. doi:10.4088/JCP.22r14772
2. Johnstone JM, Hatsu I, Tost G, et al. Micronutrients for Attention-Deficit/Hyperactivity Disorder in Youths: A Placebo-Controlled Randomized Clinical Trial. *J Am Acad Child Adolesc Psychiatry*. 2022;61(5):647-661.
3. Rucklidge JJ, Frampton CM, Gorman B, Boggis A. Vitamin-mineral treatment of attention-deficit hyperactivity disorder in adults: double-blind randomised placebo-controlled trial. *Br J Psychiatry*. 2014;204:306-315. doi:10.1192/bjp.bp.113.132126
4. Rucklidge JJ, Eggleston MJF, Johnstone JM, Darling K, Frampton CM. Vitamin-mineral treatment improves aggression and emotional regulation in children with ADHD: a fully blinded, randomised, placebo-controlled trial. *J Child Psychol Psychiatry*. 2018;59(3):232-246. doi:10.1111/jcpp.12817

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Educational content only. Not medical advice. Not a diagnosis, and not a substitute for your clinician. Never start, stop, or change ADHD medication or any other prescription based on this guide. In crisis, call or text 988 now.